

Network Reconnaissance & Vulnerability Assessment

Target Scope: Home Network Lab (192.168.1.0/24)

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Primary Toolset: Nmap v7.99

1. Executive Summary

A targeted network reconnaissance and vulnerability assessment was conducted against an isolated home lab environment (192.168.1.0/24) to evaluate its perimeter security posture and model potential attacker entry points.

Across the /24 subnet, **8 active hosts** were identified. While host isolation across most devices met baseline expectations, the assessment uncovered significant exposure on the primary edge router (192.168.1.1). Most notably, an active **Telnet service (Port 23)** is exposed, transmitting network management traffic and administrative credentials in plain text. Compounding this risk, fingerprinting indicates the router is operating on an unpatched, end-of-life Linux kernel.

Addressing the cleartext administration protocol and modernizing the perimeter device's firmware represent the highest-priority remediation items.

Risk Overview

Target IP	Severity	Finding Summary	Primary Risk
192.168.1.1	CRITICAL	Exposed Telnet Service (Port 23)	Cleartext credential interception via network sniffing
192.168.1.1	HIGH	Outdated Linux Kernel (v3.10–4.11)	Exposure to public privilege escalation & RCE exploits
192.168.1.1	MEDIUM	Non-Standard SSL Validity (10-Year Term)	Violation of TLS hygiene standards; long-term key exposure
192.168.1.1	INFORMATIONAL	Secure UDP Configuration (UPnP/NetBIOS Closed)	Reduced surface against automated internal port mapping

2. Scope & Authorization

- **Target Range:** 192.168.1.0/24 (Local Subnet)
- **Primary Scope Focus:** 192.168.1.1 (Fiberhome ISP Gateway)
- **Secondary Scope Focus:** 192.168.1.2
- **Authorization Context:** All testing was restricted strictly to self-owned, lab-isolated equipment for skill development and portfolio verification. No third-party or production systems were included in the scanning boundary.

3. Methodology & Technical Execution

The assessment followed a structured, non-destructive discovery process using **Nmap v7.99** to map the network surface, identify active listener ports, and evaluate software versions:

1. **Active Host Discovery:** Executed an ICMP/TCP ping sweep (-sn) across 192.168.1.0/24 to map active endpoints without triggering intrusive port sweeps on quiet hosts.
2. **Full TCP Port & Service Enumeration:** Scanned all 65,535 TCP ports (-sV -p-) on primary targets to confirm listening services and exact banner versions.
3. **OS Stack Fingerprinting:** Used TCP/IP stack behavior analysis (-O) to deduce the underlying operating system and kernel series running on the primary gateway.
4. **Targeted UDP Sampling:** Evaluated high-risk UDP services (-sU -p 53,67,68,137,1900) including DNS, DHCP, NetBIOS, and UPnP.
5. **Vulnerability Scripting:** Deployed select Nmap Scripting Engine (NSE) scripts (--script vuln,auth,default) to check for known default credentials and protocol-level weaknesses.

4. Detailed Audit Findings

Finding 1: Unencrypted Telnet Service Active on Gateway

- **Severity:** ● CRITICAL
- **Target Endpoint:** 192.168.1.1:23 (TCP)
- **Impact Area:** Confidentiality & Authentication Integrity

Technical Description

Service scanning confirmed an active Telnet daemon accepting connections on port 23. NSE checks confirmed a live interactive login banner. Because Telnet lacks native transport encryption, all session traffic—including administrative usernames and passwords—is sent across the local segment in plain text.

Risk Impact: An attacker gaining access to the local network (via Wi-Fi compromise or rogue device placement) can easily capture administrative session credentials using basic packet analyzer tools (e.g., Wireshark), leading to complete gateway takeovers.

PORT STATE SERVICE VERSION

23/tcp open telnet BusyBox telnetd

| telnet-encryption: Unencrypted connection established

|_Telnet banner: \r\nFiberhome Gateway Login:\r\n

Finding 2: Outdated Gateway Kernel Version

- **Severity:**  **HIGH**
- **Target Endpoint:** 192.168.1.1
- **Impact Area:** Platform Integrity & System Patch Management

Technical Description

OS stack signature matching indicates the edge router is running an embedded Linux kernel in the 3.10 to 4.11 release range. These kernel branches reached End-of-Life (EOL) between 2017 and 2019 and lack contemporary security patches.

Risk Impact: Operating obsolete kernel builds leaves the primary perimeter vulnerable to publicly available kernel exploits, potentially allowing local attackers to escalate privileges or execute arbitrary code directly on the network's main entry point.

Device type: general purpose|router

Running: Linux 3.X | 4.X

OS CPE: cpe:/o:linux:linux_kernel:3.10 cpe:/o:linux:linux_kernel:4.11

OS details: Linux 3.10 - 4.11

Finding 3: Extended SSL/TLS Certificate Validity Window

- **Severity:**  **MEDIUM**
- **Target Endpoint:** 192.168.1.1:443 (TCP)
- **Impact Area:** Cryptographic Hygiene

Technical Description

The HTTPS administration interface relies on a self-signed SSL/TLS certificate generated with a 10-year validity window (2019–2029).

Risk Impact: While functional for local encryption, multi-year certificates violate modern TLS standards (which cap validity periods at 398 days to enforce key rotation). Excessive validity windows increase the window of exposure if private keys are leaked or compromised.

443/tcp open ssl/http

| ssl-cert: Subject: commonName=Fiberhome/organizationName=ISP

| Not valid before: 2019-01-01T00:00:00

|_Not valid after: 2029-01-01T00:00:00

Finding 4: Hardened UDP Profile (UPnP & NetBIOS Disabled)

- **Severity:** ● INFORMATIONAL
- **Target Endpoint:** 192.168.1.1
- **Impact Area:** Surface Reduction (Positive Security Control)

Technical Description

Targeted UDP scanning confirmed that legacy and potentially hazardous services—specifically NetBIOS (137/udp) and Universal Plug and Play (1900/udp)—are closed.

Security Value: Disabling UPnP prevents compromised internal hosts from dynamically requesting external port forwarding rules without administrator consent.

5. Strategic Remediation Plan

1. Immediate Actions (Perimeter Hardening)

- **Disable Telnet:** Access the router management interface and turn off the Telnet daemon completely.
- **Enforce Secure Management Protocols:** Restrict command-line management to **SSH (Port 22)** and ensure web access strictly enforces **HTTPS (Port 443)** over encrypted sessions.

2. Short-Term Actions (Patching & Cryptography)

- **Update Gateway Firmware:** Check with the vendor/ISP for official firmware releases containing upgraded kernel builds.
- **Regenerate TLS Certificates:** Replace 10-year certificates with shorter-lived certificates (1-year maximum) to align with current cryptographic standards.

3. Long-Term Actions (Architecture)

- **Hardware Lifecycle Management:** If the ISP or vendor no longer provides active security maintenance or kernel updates for the Fiberhome device, replace the gateway hardware.
- **Subnet Segmentation:** Isolate administrative interfaces from general user/guest Wi-Fi traffic using VLANs or dedicated management subnets.

6. Conclusion

The assessment verified that while fundamental perimeter defenses (such as closing unnecessary UDP services) are present, critical weaknesses exist on the main edge router. The combination of plain-text management protocols (Telnet) and an obsolete kernel leaves the network perimeter exposed to internal credential harvesting and known kernel exploits.

Executing the Phase 1 and Phase 2 remediation recommendations will eliminate cleartext credential exposure and restore perimeter hygiene to modern standards.